

05_LORA_GATEWAY_RESULTS

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EVIDENCE PROVENANCE: [FAILOVER BENCH TEST]

Empirical validation of emergency 1.0 Hz Safe Beacon emission and gateway capture upon Wi-Fi infrastructure loss.

1. Safe Beacon Autonomous Failover Mechanics

When primary Wi-Fi drops for >1000 ms, vehicles autonomously switch to emergency 433 MHz LoRa beaconing at 1.0 Hz (BEACON,ID,seq,speed,status). The host gateway receives this beacon, flags COMM_DEGRADED on the Twin, and clamps vehicle safe speed to crawl.

2. Failover Test Results (50 Trials)

Failover Event	Observed Timing	Safety Standard	Result
Wi-Fi Loss Detection	1000 ms	1000 ms timeout window	PASS
Beacon Trigger Delay	24.5 ms after loss	< 50.0 ms trigger budget	PASS
First Beacon Ingested	1063 ms total	< 1500 ms safe window	PASS
Twin Status Transition	12.0 ms post RX	< 50.0 ms twin update	PASS
Control Room Alert Render	18.5 ms post twin	< 50.0 ms UI render	PASS
Beacon Capture Rate	100% (50/50 trials)	Zero lost failover events	PASS